CS 401

FINALTERM PREPARATION (2026)

LECTURE WISE MCQS

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Chapter 10 (Debug Interrupts)

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MCQ 1

Which interrupt is generated after every instruction when Trap Flag is set?

A) INT 0
B) INT 1
C) INT 2
D) INT 3

✅ **Correct Option:** B

Answer: INT 1

Explanation:
When Trap Flag (TF) is set to 1, the CPU generates INT 1 after executing every instruction.

MCQ 2

Trap Flag is part of which register?

A) AX
B) IP

C) FLAGS
D) SP

✅ **Correct Option:** C

Answer: FLAGS

Explanation:
Trap Flag is a control bit located in the FLAGS register.

MCQ 3

INT 3 is mainly used for:

A) Arithmetic operations
B) Memory allocation
C) Breakpoint debugging
D) Program termination

✅ **Correct Option:** C

Answer: Breakpoint debugging

Explanation:
INT 3 stops program execution at a specific point for debugging.

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MCQ 4

What happens when execution reaches an INT 3 instruction?

A) Program restarts
B) Program terminates
C) Control transfers to debugger
D) Instruction is ignored

✅ **Correct Option:** C

Answer: Control transfers to debugger

Explanation:
INT 3 halts execution and passes control to the debugger.

MCQ 5

Opcode of INT 3 is:

A) CD
B) C1
C) CC
D) C9

✅ **Correct Option:** C

Answer: CC

Explanation:
INT 3 uses the one-byte opcode CC (hex).

MCQ 6

Which debugging method is slower but more detailed?

A) Breakpoint debugging
B) Single step debugging
C) Hardware debugging
D) Batch debugging

✅ **Correct Option:** B

Answer: Single step debugging

Explanation:
Single step debugging executes one instruction at a time, making it slow but detailed.

MCQ 7

Single step debugging allows the programmer to:

A) Skip instructions
B) Execute multiple instructions at once

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C) Execute one instruction at a time
D) Modify source code automatically

✅ **Correct Option:** C

Answer: Execute one instruction at a time

Explanation:
INT 1 enables instruction-by-instruction execution.

MCQ 8

Which interrupt does NOT require insertion into program code?

A) INT 1
B) INT 3
C) Both INT 1 and INT 3
D) None

✅ **Correct Option:** A

Answer: INT 1

Explanation:
INT 1 is generated automatically when Trap Flag is set; it is not inserted in code.

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MCQ 9

Breakpoint debugging is best used when:

A) Every instruction must be checked
B) Program needs to stop at a specific location
C) Program must run slowly
D) Flags are not required

✅ **Correct Option:** B

Answer: Program needs to stop at a specific location

Explanation:
INT 3 stops execution only at defined breakpoints.

MCQ 10

Which register controls single step execution indirectly?

A) AX
B) IP
C) FLAGS
D) CS

✅ **Correct Option:** C

Answer: FLAGS

Explanation:
Trap Flag in FLAGS register controls single step execution.

MCQ 11

Which of the following is TRUE about INT 1?

A) It stops program permanently
B) It is used for program termination
C) It occurs after each instruction
D) It is faster than INT 3

✅ **Correct Option:** C

Answer: It occurs after each instruction

Explanation:
INT 1 is generated after every instruction when TF is enabled.

MCQ 12

Which debugging interrupt uses a special one-byte instruction?

A) INT 0
B) INT 1

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C) INT 2
D) INT 3

✅ **Correct Option:** D

Answer: INT 3

Explanation:
INT 3 uses a one-byte opcode CC, making it ideal for breakpoints.

MCQ 13

Why is debugging important in assembly language?

A) Assembly has automatic error handling
B) Assembly programs are very long
C) Small mistakes can cause serious errors
D) Assembly does not use registers

✅ **Correct Option:** C

Answer: Small mistakes can cause serious errors

Explanation:
Assembly is low-level; even one wrong instruction can crash the program.

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MCQ 14

Which debugging method allows normal program execution until a stop point?

A) Single step debugging
B) Breakpoint debugging
C) Instruction decoding
D) Memory mapping

✅ **Correct Option:** B

Answer: Breakpoint debugging

Explanation:
Breakpoint debugging allows normal execution until INT 3 is reached.

MCQ 15

Debug interrupts are mainly used to:

A) Increase execution speed
B) Control and analyze program execution
C) Allocate memory
D) Handle input/output

✅ **Correct Option:** B

Answer: Control and analyze program execution

Explanation:
Debug interrupts help observe registers, flags, memory, and execution flow.

3 Important Short Questions (with Answers)

Q1: What is debugging?

Answer:
Debugging is the process of detecting, analyzing, and correcting errors in a program. It helps programmers understand how a program executes and where it fails.

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Q2: What is Trap Flag and where is it located?

Answer:
Trap Flag (TF) is a control bit located in the **FLAGS register**. When TF is set to 1, the processor generates INT 1 after every instruction.

Q3: Write difference between INT 1 and INT 3.

Answer:
INT 1 executes the program instruction by instruction using Trap Flag, while INT 3 stops the program at a specific location using a breakpoint instruction.

1 Important Long Question (Fully Explained)

Q: What are Debug Interrupts? Explain INT 1 and INT 3 in detail.

Answer:

Debug Interrupts are special interrupts used to analyze and control program execution during debugging. They help programmers detect logical errors by monitoring registers, flags, memory, and instruction flow.

INT 1 (Single Step Interrupt):

INT 1 allows execution of a program one instruction at a time. It works through the Trap Flag present in the FLAGS register. When Trap Flag is enabled, the CPU generates an interrupt after executing each instruction. This interrupt transfers control to the debugger, allowing detailed inspection of program behavior. INT 1 is very useful for deep analysis but is slow.

INT 3 (Breakpoint Interrupt):

INT 3 is used to stop the program at a specific location. Its opcode is CC. A debugger inserts INT 3 at a desired instruction. When the program reaches this instruction, execution halts and control is given to the debugger. INT 3 is faster and efficient for checking loops and specific code segments.

Conclusion:

Both INT 1 and INT 3 are powerful debugging tools. INT 1 is used for detailed instruction-level debugging, while INT 3 is used for fast breakpoint-based debugging.

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Chapter 11: Multitasking

✅ 15 MCQs (with Answer & Explanation)

MCQ 1

Multitasking means:

A) Executing one program at a time
B) Executing multiple programs simultaneously
C) Writing multiple programs
D) Compiling multiple programs

✅ **Correct Option:** B
Answer: Executing multiple programs simultaneously

Explanation:
Multitasking allows more than one task to appear to run at the same time by sharing CPU time.

MCQ 2

A task in multitasking refers to:

A) A register
B) A memory location
C) A running program
D) An interrupt

✅ **Correct Option:** C
Answer: A running program

Explanation:
A task is a program or process currently being executed by the CPU.

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MCQ 3

In multitasking, the CPU actually executes:

A) Many tasks at the same time
B) Only one task at a time
C) Two tasks at a time
D) No task at a time

✅ **Correct Option:** B
Answer: Only one task at a time

Explanation:
CPU switches very fast between tasks, creating the illusion of parallel execution.

MCQ 4

Context switching means:

A) Switching memory
B) Switching instructions
C) Saving and restoring task state
D) Restarting a program

✅ **Correct Option:** C
Answer: Saving and restoring task state

Explanation:
Context switching saves the current task's state and loads the next task's state.

MCQ 5

Which of the following is saved during context switching?

A) Screen output
B) Registers and flags
C) Source code
D) Compiler data

✅ **Correct Option:** B
Answer: Registers and flags

Explanation:
Registers, flags, stack, and instruction pointer are saved.

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MCQ 6

Time slicing means:

A) Dividing memory
B) Dividing CPU time among tasks
C) Dividing hard disk
D) Dividing stack

✅ **Correct Option:** B
Answer: Dividing CPU time among tasks

Explanation:
Each task gets a small portion of CPU time.

MCQ 7

Which interrupt is commonly used for task switching?

A) Keyboard interrupt
B) Timer interrupt
C) Disk interrupt
D) Printer interrupt

✅ **Correct Option:** B
Answer: Timer interrupt

Explanation:
Timer interrupts occur periodically and trigger context switching.

MCQ 8

Elaborate multitasking improves:

A) Program size
B) CPU utilization
C) Program errors
D) Memory wastage

✅ **Correct Option:** B
Answer: CPU utilization

Explanation:
CPU remains busy by switching between tasks.

MCQ 9

TSR stands for:

A) Task System Routine
B) Temporary System Register
C) Terminate and Stay Resident
D) Task Sharing Register

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✅ **Correct Option:** C
Answer: Terminate and Stay Resident

Explanation:
TSR programs remain in memory after execution.

MCQ 10

A TSR program runs:

A) Only once
B) In foreground
C) In background
D) Only at boot time

✅ **Correct Option:** C
Answer: In background

Explanation:
TSR stays resident and works in the background.

MCQ 11

Multitasking kernel as TSR means:

A) Kernel runs once
B) Kernel stays in memory
C) Kernel is removed after execution
D) Kernel uses disk memory

✅ **Correct Option:** B
Answer: Kernel stays in memory

Explanation:
Kernel remains resident to manage multiple tasks.

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✅ 3 Important Short Questions (with Proper Answers)

Q1: What is multitasking?

Answer:
Multitasking is the ability of a system to execute multiple tasks by sharing CPU time. The processor switches between tasks so quickly that they appear to run simultaneously.

Q2: What is context switching?

Answer:
Context switching is the process of saving the current task's state (registers, flags, stack, and instruction pointer) and loading another task's state to resume its execution.

Q3: What is a TSR program?

Answer:
A TSR (Terminate and Stay Resident) program remains in memory after execution and runs in the background to provide services when needed.

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✔ 1 Important Long Question (Exam-Ready)

Q: Explain multitasking and describe how a multitasking kernel works as a TSR.

Answer:

Multitasking is the technique that allows multiple programs, called tasks, to share CPU time. Although the CPU can execute only one task at a time, rapid switching between tasks creates the illusion of simultaneous execution.

Multitasking is implemented using **context switching**. In this process, the state of the currently running task is saved, and the state of another task is loaded. CPU time is divided into small intervals known as **time slices**, and each task is given a time slice for execution. Timer interrupts are commonly used to trigger task switching.

A multitasking kernel can be implemented as a **TSR (Terminate and Stay Resident)** program. The kernel stays resident in memory after execution and continues to manage multiple tasks in the background. It handles task scheduling, context switching, and interrupt handling. TSR-based

multitasking is especially useful in DOS-based systems where full operating system support is not available.

Thus, multitasking improves CPU utilization, system efficiency, and responsiveness.

Chapter 12 (Video Services)

15 MCQs (with Answer & Explanation)

MCQ 1

Video services are mainly used to:

A) Manage memory
B) Control screen output
C) Access disk
D) Handle files

✅ **Correct Answer:** B

Explanation:
Video services handle screen display such as text, graphics, cursor movement, and screen clearing.

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MCQ 2

Which interrupt is used for BIOS video services?

A) INT 13H
B) INT 21H
C) INT 10H
D) INT 16H

✅ **Correct Answer:** C

Explanation:
BIOS video services are accessed using interrupt **INT 10H**.

MCQ 3

BIOS video services are provided by:

A) DOS
B) Operating System
C) Hardware
D) BIOS

✅ **Correct Answer:** D

Explanation:
BIOS video services are part of the BIOS firmware.

MCQ 4

Which of the following supports graphics mode?

A) DOS video services
B) BIOS video services
C) File services
D) Keyboard services

✅ **Correct Answer:** B

Explanation:
Only BIOS video services support graphics mode.

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MCQ 5

DOS video services are accessed using which interrupt?

A) INT 10H
B) INT 21H
C) INT 13H
D) INT 17H

✅ **Correct Answer:** B

Explanation:
DOS services, including video services, use **INT 21H**.

MCQ 6

Which video service level is closer to hardware?

A) DOS
B) BIOS
C) Application
D) Compiler

✅ **Correct Answer:** B

Explanation:
BIOS services directly interact with hardware.

MCQ 7

Which service is easier for simple text output?

A) BIOS video services
B) DOS video services
C) Hardware interrupts
D) Memory services

✅ **Correct Answer:** B

Explanation:
DOS video services are high-level and easier to use for text output.

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MCQ 8

INT 10H is used to:

A) Print files
B) Manage memory
C) Handle video operations
D) Read keyboard

✅ **Correct Answer:** C

Explanation:
INT 10H provides BIOS video-related functions.

MCQ 9

Which function clears the screen using BIOS services?

A) Keyboard interrupt
B) Resetting video mode
C) Disk interrupt
D) Timer interrupt

✅ **Correct Answer:** B

Explanation:
Resetting the video mode using INT 10H clears the screen.

MCQ 10

Which register usually holds BIOS function number?

A) AL
B) BX
C) AH
D) DL

✅ **Correct Answer:** C

Explanation:
BIOS function numbers are passed through the **AH** register.

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✅ 3 Short Questions (with Proper Answers)

Q1: What are video services?

Answer:
Video services are system services used to display text or graphics on the screen, move the cursor, clear the screen, and control video modes.

Q2: What are BIOS video services?

Answer:
BIOS video services are low-level services provided by the BIOS and accessed through INT 10H to control screen display and graphics.

Q3: What are DOS video services?

Answer:
DOS video services are high-level services provided by DOS and accessed using INT 21H for simple text output and screen operations.

✔ 1 Long Exam Question (Fully Explained)

Q: Explain Video Services. Differentiate between BIOS Video Services and DOS Video Services.

Answer:

Video services are system-level services used to control screen display, including text output, graphics, cursor movement, and screen clearing. In assembly language, these services are accessed using software interrupts instead of directly accessing video hardware.

BIOS Video Services are provided by the BIOS and accessed through interrupt INT 10H. These services operate at a low level and allow direct control over video hardware. BIOS video services support both text and graphics modes, are faster, and can work without the presence of DOS.

DOS Video Services are provided by the DOS operating system and accessed using interrupt INT 21H. These services are high-level, easier to use, and mainly support text output. DOS video services depend on DOS and are slower compared to BIOS services.

In conclusion, BIOS video services are suitable for low-level and graphics-related operations, while DOS video services are suitable for simple text-based applications.

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Chapter 13: Secondary Storage

MCQ 1

Secondary storage is used to:

A) Store data temporarily
B) Store data permanently
C) Execute programs
D) Increase CPU speed

✅ **Correct Answer:** B

Explanation:
Secondary storage is non-volatile, meaning data is not lost when power is off.

MCQ 2

Which of the following is an example of secondary storage?

A) RAM
B) Cache
C) Hard Disk
D) Register

✅ **Correct Answer:** C

Explanation:
Hard disk stores data permanently and is a secondary storage device.

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MCQ 3

Data on a hard disk is physically stored in the form of:

A) Files only
B) Registers
C) Tracks and sectors
D) Pages

✅ **Correct Answer:** C

Explanation:
Hard disks store data in tracks, sectors, and cylinders.

MCQ 4

The smallest addressable unit on a disk is called:

A) Track
B) Cylinder
C) Sector
D) Platter

✅ **Correct Answer:** C

Explanation:
A sector is the smallest unit that can be read or written.

MCQ 5

Which interrupt is used for BIOS disk services?

A) INT 10H
B) INT 21H
C) INT 13H
D) INT 16H

✅ **Correct Answer:** C

Explanation:
INT 13H provides BIOS-level disk access.

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MCQ 6

BIOS storage access is considered:

A) High-level
B) Application level
C) Low-level
D) User level

✅ **Correct Answer:** C

Explanation:
BIOS access directly communicates with hardware.

MCQ 7

Which of the following requires knowledge of disk geometry?

A) DOS disk access
B) BIOS disk access
C) File system
D) Device drivers

✅ **Correct Answer:** B

Explanation:
BIOS access needs cylinder, head, and sector information.

MCQ 8

DOS storage services are accessed using:

A) INT 10H
B) INT 13H
C) INT 21H
D) INT 16H

✅ **Correct Answer:** C

Explanation:
INT 21H provides DOS file and disk services.

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MCQ 9

DOS disk access works with:

A) Sectors
B) Cylinders
C) Files
D) Tracks

✅ **Correct Answer:** C

Explanation:
DOS accesses storage using files, not raw sectors.

MCQ 10

Which method is easier for programmers?

A) BIOS disk access
B) DOS disk access
C) Direct hardware access
D) Register access

✅ **Correct Answer:** B

Explanation:
DOS handles buffering and errors automatically.

✅ 3 Short Questions (with Proper Answers)

Q1: What is secondary storage?

Answer:
Secondary storage is non-volatile storage used to store data permanently, even when the computer is turned off.

Q2: What is BIOS storage access?

Answer:
BIOS storage access is low-level disk access provided through INT 13H, allowing direct communication with storage hardware.

Q3: What is a device driver?

Answer:
A device driver is a program that allows the operating system to communicate with and control a specific hardware device.

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✅ 1 Long Exam Question (Fully Explained)

Q: Explain secondary storage. Describe storage access using BIOS, storage access using DOS, and the role of device drivers.

Answer:

Secondary storage is used to store data permanently and is non-volatile in nature. Common secondary storage devices include hard disks, USB drives, and optical disks. Data is physically stored in the form of tracks, sectors, and cylinders on disk platters.

Storage access using BIOS is performed through interrupt INT 13H. BIOS provides low-level access to disk hardware and requires detailed knowledge of disk geometry such as cylinder, head, and sector. BIOS access is fast and works without DOS, but it is complex and hardware-dependent.

Storage access using DOS is performed through interrupt INT 21H. DOS provides high-level disk access using files instead of raw sectors. It is easier to use, portable, and includes automatic error handling, but it is slower and depends on DOS.

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Lecture 14: Serial Port Programming

MCQ 1

Serial communication transmits data:

A) Byte by byte
B) Bit by bit
C) Word by word
D) Block by block

✅ **Correct Answer:** B

Explanation:
In serial communication, data is sent one bit at a time over a single line.

MCQ 2

Which device controls serial communication in a PC?

A) ALU
B) CPU
C) UART
D) DMA

✅ **Correct Answer:** C

Explanation:
UART (Universal Asynchronous Receiver Transmitter) handles serial data transmission and reception.

MCQ 3

Which of the following is an application of serial communication?

A) RAM access
B) Keyboard input
C) Modem communication
D) Cache memory

✅ **Correct Answer:** C

Explanation:
Serial communication is widely used in modem and device communication.

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MCQ 4

Which type of serial communication does NOT use a clock signal?

A) Synchronous
B) Parallel
C) Asynchronous
D) Optical

✅ **Correct Answer:** C

Explanation:
Asynchronous communication does not use a shared clock.

MCQ 5

Start and stop bits are used in:

A) Parallel communication
B) Synchronous communication
C) Asynchronous serial communication
D) Optical communication

✅ **Correct Answer:** C

Explanation:
Asynchronous serial communication uses start and stop bits to frame data.

MCQ 6

Baud rate refers to:

A) Number of characters per second
B) Number of bits per second
C) Number of bytes per second
D) Clock speed

✅ **Correct Answer:** B

Explanation:
Baud rate defines the number of bits transmitted per second.

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MCQ 7

Which of the following is NOT part of a serial data frame?

A) Start bit
B) Data bits
C) Parity bit
D) Address bus

✅ **Correct Answer:** D

Explanation:
Address bus is not related to serial communication frames.

MCQ 8

COM1 and COM2 are examples of:

A) Parallel ports
B) Serial ports
C) USB ports
D) Network ports

✅ **Correct Answer:** B

Explanation:
COM ports are serial communication ports.

MCQ 9

Which communication method requires fewer wires?

A) Parallel
B) Serial
C) Optical
D) Wireless

✅ **Correct Answer:** B

Explanation:
Serial communication uses fewer wires than parallel communication.

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MCQ 10

Which type of communication is better for long distances?

A) Parallel
B) Serial
C) Cache
D) Memory

✅ **Correct Answer:** B

Explanation:
Serial communication is reliable over long distances.

✔ 3 Short Questions (with Proper Answers)

Q1: What is serial communication?

Answer:
Serial communication is a method of data transmission in which data is sent one bit at a time over a single communication line.

Q2: What is UART?

Answer:
UART (Universal Asynchronous Receiver Transmitter) is a hardware device that converts parallel data from the CPU into serial form and vice versa.

Q3: What is baud rate?

Answer:
Baud rate is the speed of serial communication and represents the number of bits transmitted per second.

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✔ 1 Long Exam Question (Fully Explained)

Q: Explain serial port programming and describe serial communication in detail.

Answer:

Serial port programming is the process of controlling serial ports to transmit and receive data between a computer and external devices. It allows low-level communication by sending data one bit at a time.

Serial communication is a data transmission technique in which bits are transmitted sequentially over a single line. It can be synchronous or asynchronous. Asynchronous communication does not use a common clock and instead uses start and stop bits to synchronize data transfer. A typical serial data frame includes a start bit, data bits, an optional parity bit for error detection, and one or more stop bits.

Lecture 15: Protected Mode Programming

MCQ 1

Protected mode was introduced in which processor?

A) 8086
B) 80186
C) 80286
D) 80386

✅ **Correct Answer:** C

Explanation:
Protected mode was first introduced with the Intel 80286 processor.

MCQ 2

Which problem of real mode is solved by protected mode?

A) Low speed
B) Limited memory access
C) Lack of registers
D) Lack of instructions

✅ **Correct Answer:** B

Explanation:
Real mode can access only 1 MB memory, while protected mode supports much more.

MCQ 3

32-bit programming allows access to memory up to:

A) 1 MB
B) 16 MB
C) 64 MB
D) 4 GB

✅ **Correct Answer:** D

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Explanation:
32-bit addressing allows access to 4 GB memory space.

MCQ 4

Which of the following is a 32-bit register?

A) AX
B) BX
C) EAX
D) CX

✅ **Correct Answer:** C

Explanation:
Registers starting with "E" are 32-bit registers.

MCQ 5

What does the prefix "E" in EAX stand for?

A) Extra
B) Enhanced
C) Extended
D) Embedded

✅ **Correct Answer:** C

Explanation:
"E" stands for Extended, indicating 32-bit registers.

MCQ 6

VESA stands for:

A) Video Enhanced System Architecture
B) Video Electronics Standards Association
C) Visual Enhanced System Adapter
D) Virtual Electronic Screen Access

✅ **Correct Answer:** B

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Explanation:
VESA is an organization that defines graphics standards.

MCQ 7

Linear Frame Buffer is used for:

A) Keyboard input
B) File storage
C) Graphics display
D) Interrupt handling

✅ **Correct Answer:** C

Explanation:
Linear Frame Buffer maps memory directly to the screen for graphics.

MCQ 8

Main advantage of Linear Frame Buffer is:

A) Less memory usage
B) Bank switching required
C) Direct memory access to pixels
D) Slower performance

✅ **Correct Answer:** C

Explanation:
Each pixel corresponds to a memory address, making graphics programming easier.

MCQ 9

Interrupt handling in protected mode uses:

A) IVT
B) IDT
C) GDT
D) LDT

✅ **Correct Answer:** B

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Explanation:
Protected mode uses the Interrupt Descriptor Table (IDT).

MCQ 10

Which table replaces IVT in protected mode?

A) GDT
B) LDT
C) IDT
D) TSS

✅ **Correct Answer:** C

Explanation:
IDT replaces the Interrupt Vector Table used in real mode.

✅ 3 Short Questions (with Proper Answers)

Q1: What is protected mode?

Answer:
Protected mode is an operating mode of x86 processors that provides large memory access, memory protection, and multitasking capabilities.

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Q2: What is Linear Frame Buffer?

Answer:
Linear Frame Buffer is a continuous block of memory mapped directly to video memory, where each memory location represents a pixel on the screen.

Q3: What is IDT?

Answer:
IDT (Interrupt Descriptor Table) is a table used in protected mode to store addresses of interrupt service routines.

✅ 1 Long Exam Question (Fully Explained)

Q: Explain Protected Mode Programming. Discuss 32-bit programming, VESA Linear Frame Buffer, and interrupt handling in protected mode.

Answer:

Protected mode is an advanced operating mode of x86 processors introduced to overcome the limitations of real mode. It allows access to large memory, provides memory protection, and supports multitasking, making modern operating systems stable and secure.

In protected mode, 32-bit programming is used, which allows the processor to handle 32-bit data and access up to 4 GB of memory. Extended registers such as EAX, EBX, and ECX are used for faster and efficient processing.

VESA Linear Frame Buffer is a graphics technique that maps video memory as a continuous memory block. It simplifies graphics programming by allowing direct access to pixels without bank switching.

Interrupt handling in protected mode is managed using the Interrupt Descriptor Table (IDT). The IDT stores addresses of interrupt service routines and provides controlled and secure handling of hardware and software interrupts.

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Lecture 16: Interfacing with High Level Languages

MCQ 1

Interfacing with High Level Languages means:

A) Writing only assembly programs
B) Writing only C programs
C) Using assembly and high-level languages together
D) Using machine language

✅ **Correct Answer:** C

Explanation:
Interfacing allows assembly and high-level languages like C to work together.

MCQ 2

Which high-level language is most commonly interfaced with assembly?

A) Java
B) Python
C) C
D) PHP

✅ **Correct Answer:** C

Explanation:
C is commonly interfaced with assembly because of its close relation to hardware.

MCQ 3

Calling conventions define:

A) CPU speed
B) Memory size
C) Rules for function calls
D) Instruction formats

✅ **Correct Answer:** C

Explanation:
Calling conventions define how arguments, return values, and stack cleanup are handled.

MCQ 4

In C calling convention, function arguments are passed through:

A) Registers
B) Heap
C) Stack
D) Cache

✅ **Correct Answer:** C

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Explanation:
Arguments are pushed onto the stack.

MCQ 5

In C calling convention, arguments are pushed on the stack in which order?

A) Left to right
B) Right to left
C) Random order
D) Alphabetical order

✅ **Correct Answer:** B

Explanation:
C pushes function arguments from right to left.

MCQ 6

Who is responsible for cleaning the stack in C calling convention?

A) Callee
B) Compiler
C) Caller
D) Operating system

✅ **Correct Answer:** C

Explanation:
The caller removes arguments from the stack.

MCQ 7

Return value of a function is usually stored in:

A) BX
B) CX
C) DX
D) AX / EAX

✅ **Correct Answer:** D

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Explanation:
AX (16-bit) or EAX (32-bit) is used to return values.

MCQ 8

Which keyword is used in assembly to declare a C function?

A) PUBLIC
B) EXTERN
C) GLOBAL
D) DEFINE

✅ **Correct Answer:** B

Explanation:
EXTERN is used to access C functions in assembly.

MCQ 9

Calling C from Assembly means:

A) Writing both codes in C
B) Calling assembly functions in C
C) Calling C functions inside assembly code
D) Compiling assembly as C

✅ **Correct Answer:** C

Explanation:
Assembly program calls a function written in C.

MCQ 10

Calling Assembly from C requires assembly function to follow:

A) Java convention
B) Pascal convention
C) C calling convention
D) BIOS rules

✅ **Correct Answer:** C

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Explanation:
Assembly functions must follow C calling conventions.

✔ 3 Short Questions (with Proper Answers)

Q1: What are calling conventions?

Answer:
Calling conventions are a set of rules that define how function parameters are passed, how return values are returned, and how the stack is managed.

Q2: Why is assembly language interfaced with C?

Answer:
Assembly is interfaced with C to improve performance and allow direct access to hardware.

Q3: What is the role of the stack in interfacing?

Answer:
The stack is used to pass function arguments and store return addresses during function calls.

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✔ 1 Long Exam Question (Fully Explained)

Q: Explain interfacing with high-level languages. Discuss calling conventions, calling C from assembly, and calling assembly from C.

Answer:

Interfacing with high-level languages means combining assembly language with languages such as C to take advantage of both low-level hardware control and high-level programming simplicity.

Calling conventions define the rules that both assembly and C must follow for correct function calls. These rules specify how arguments are passed, how return values are returned, which

registers are used, and who cleans the stack. In C calling convention, arguments are pushed onto the stack from right to left, and the caller cleans the stack.

Calling C from assembly involves pushing function arguments onto the stack and using the CALL instruction to invoke a C function. The return value is received in AX or EAX.

Calling assembly from C involves writing an assembly function that follows the C calling convention. The C program calls the assembly function like a normal C function, and the assembly code accesses parameters from the stack and returns the result through AX or EAX.

Lecture 17: Comparison with Other Processors

MCQ 1

Motorola 68000 processor belongs to which architecture?

A) RISC
B) CISC
C) VLIW
D) EPIC

✅ **Correct Answer:** B

Explanation:
Motorola 68000 is a Complex Instruction Set Computer (CISC) processor.

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MCQ 2

Sun SPARC processor follows which architecture?

A) CISC
B) RISC
C) SIMD
D) MIMD

✅ **Correct Answer:** B

Explanation:
SPARC is based on Reduced Instruction Set Computer (RISC) architecture.

MCQ 3

Motorola 68K has how many data registers?

A) 4
B) 6
C) 8
D) 16

✅ **Correct Answer:** C

Explanation:
Motorola 68K has 8 data registers (D0–D7).

MCQ 4

Which registers are used as address registers in 68K?

A) D0–D7
B) A0–A7
C) R0–R7
D) EAX–EDI

✅ **Correct Answer:** B

Explanation:
A0–A7 are address registers in the Motorola 68K processor.

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MCQ 5

Which register acts as the stack pointer in Motorola 68K?

A) D7
B) A5
C) A6
D) A7

✅ **Correct Answer:** D

Explanation:
A7 is used as the stack pointer in 68K architecture.

MCQ 6

Which processor uses register windows?

A) x86
B) Motorola 68K
C) Sun SPARC
D) ARM

✅ **Correct Answer:** C

Explanation:
SPARC uses register windows to speed up function calls.

MCQ 7

Main advantage of RISC processors is:

A) Complex instructions
B) Large instruction size
C) Fast execution
D) Segmented memory

✅ **Correct Answer:** C

Explanation:
RISC processors execute simple instructions quickly.

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MCQ 8

Which processor uses a flat memory model?

A) x86 only
B) Motorola 68K
C) SPARC only
D) Both 68K and SPARC

✅ **Correct Answer:** D

Explanation:
Both Motorola 68K and SPARC use a flat memory model.

MCQ 9

Which processor is widely used in servers and workstations?

A) Motorola 68K
B) x86
C) Sun SPARC
D) 8086

✅ **Correct Answer:** C

Explanation:
SPARC processors are commonly used in servers and workstations.

MCQ 10

Which architecture uses load-store design?

A) CISC
B) RISC
C) VLIW
D) SISD

✅ **Correct Answer:** B

Explanation:
RISC processors like SPARC use load-store architecture.

✅ **Correct Answer:** C

Explanation:
SPARC is designed for high-performance and scalability.

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✅ 3 Short Questions (with Proper Answers)

Q1: What type of processor is Motorola 68000?

Answer:
Motorola 68000 is a CISC-based processor with a clean and orthogonal instruction set.

Q2: What is the main feature of Sun SPARC processor?

Answer:
The main feature of Sun SPARC is the use of register windows, which improve function call performance.

Q3: What is the difference between CISC and RISC processors?

Answer:
CISC processors use complex instructions, while RISC processors use simple instructions for faster execution.

✔ 1 Long Exam Question (Fully Explained)

Q: Compare x86 processors with Motorola 68000 and Sun SPARC processors. Discuss their architectures, features, and performance.

Answer:

Different processors follow different architectural designs to achieve performance and efficiency. x86 processors are based on CISC architecture and use segmented memory. They have complex instructions and limited registers.

Motorola 68000 is also a CISC processor but has a clean and orthogonal instruction set. It uses a flat memory model and provides separate data and address registers, making it easier for compilers and programmers.

Sun SPARC is a RISC-based processor designed for high performance. It uses a load-store architecture and introduces register windows, which reduce memory access during function calls and improve execution speed. SPARC processors are widely used in servers and workstations.

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